

Manual

Archiving of Shop Floor / Order Data BDE-ARC 8.1

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1 Archiving of Business / Order Data

Purpose

The function package Archiving business /order data makes it possible to access archived data and, moreover, to export recorded data and to load these data back into the system as needed.

Implementation considerations

You use the function package Archiving business / order data, if:

- You would like to have access to data already moved to archive tables in the function package Entering business data.
- Because of legal requirements or customer demands, you need to ensure that the data entered is stored long-term.
- Because of legal requirements or customer demands, you need to import data that has already been exported back into the system to be evaluated / analyzed again.

Integration

When business / order data is archived, inventory data and recorded data is accessed from the relevant archive tables in the function package Entering business data.

Features

- Direct access to archive tables from the application
 - Direct access to archived business/order data from the applications
 - Order overview
 - Order information
 - Order shift log
 - Personnel shift log
 - Personnel report
 - Schedule controlling
 - Overhead cost controlling
 - Maintenance controlling
 - Production controlling
 - Scrap statistics
 - Scrap profile
- Transport function
 - Functions for transporting data from the on-line tables into archive tables.
- Export function

- Function used to transfer (export) data from the archive tables to external file systems for the purpose of storing the recorded business / order data long-term.
- Import function
 - Functions for importing exported data into the archive tables in order to evaluate them using the applications listed above.

2 BDE-Specific Configurations

Overview

In the BDE module, by default data are held in cache for 35 days before they are moved into long-term storage.

In a variety of BDE reports, there is the option to pull up data that are 35 days old or older. To do this, the BDE postings are set in a special medium-term or archive area. You automatically have access to such data for the most part if the selection period exceeds the short-time data area. In some applications, there is the option to "Consider long-term data" in the selection area, which can be accessed from here.

The data considered during BDE archiving include:

- Order backlog
- Order status
- Order actions
- Order sequences
- Order networks
- Lists of material components and production resources and tools
- Order serial numbers
- Additional information (long texts)
- Order quantities
- Additional order information
- Order posting records
- Order backlog (PPS)
- Additional order information (PPS)
- Order actions (PPS)
- Master detail user field (specific table)
- Order specific events
- Personal events
- Additional information for events

- Order logging information
- Additional order logging information
- BDE configuration logging information
- Additional BDE configuration logging information

Configuration

You can use HYDRA data management to configure the retention period for the data in each of the data areas.

When transferring data into the archive tables, the data for which the "retention period" (in number of days/ months/ years; see the values in parentheses) has been exceeded is transferred. If the archiving license relevant to the BDE is not available, the data will be deleted after the set retention period.

Product	Object	Object designation	Transfer	Factory default interval
BDE	ANR	Order backlog -Status, - Actions, - Sequences, - Networks, - Lists of material, - Serial numbers, - Additional information, - Quantities, - PPS data, - Master detail user fields	Online data → medium-term archive	35 days
BDE	A_ANR	Long-term archiving: Order backlog -Status, - Actions, - Sequences, - Networks, - Lists of material, - Serial numbers, - Additional information, - Quantities, - PPS data, - Master detail user fields	Medium-term archive: → long-term archive	2 years
BDE	ADEPRO	Posting records of orders	Online data → medium-term archive	35 days 1)
BDE	ADEPRO_ADD	Posting record of orders – additional information	Online data → data is deleted (it is not transferred to the medium-	Delete if the relevant posting record of the

Product	Object	Object designation	Transfer	Factory default interval
			term data)	order is no longer available
BDE	A_ADEPRO	Long-term archiving: Posting records of orders	Medium-term archive → long-term archive	2 years 1)
BDE	EREIGADEA	Order-related events incl. additional information	Online data → medium-term archive	35 days 2)
BDE	A_EREIGADEA	Long-term archiving: Order-related events incl. additional information	Medium-term archive: → long-term archive	2 years 2)
BDE	EREIGADEP	Personal events incl. additional information	Online data → medium-term archive	35 days 2)
BDE	A_EREIGADEP	Long-term archiving: Personal events incl. additional information	Medium-term archive: → long-term archive	2 years 2)
BDE	ANRLOG	HYDRA logging data	Online data → medium-term archive	35 days
BDE	A_ANRLOG	Long-term archiving HYDRA logging data	Medium-term archive: → long-term archive	3 years
BDE	CFGLOG	HYDRA logging configuration	Online data → medium-term archive	35 days
BDE	A_CFGLOG	Long-term archiving HYDRA logging configuration	Medium-term archive: → long-term archive	3 years

Please note:

- 1) If the values entered for ADEPRO or A_ADEPRO are changed (increased), the entries for ANR or A_ANR will also have to be changed (increased) accordingly.

Provided that the BDE log records are to be archived at the earliest after the OP has been archived, this can be achieved by defining the following condition for the object ADEPRO in the field of the same name within the "data management" configuration:

ade_protokoll.auftrag_nr in (select auftrag_nr from a_auftrag_status)

Please note that the [order-related postings](#) only allow for data of the online data area to be selected and edited.

- 2) Please note that the [event maintenance](#) only allows for data of the online data area to be selected and edited.

3 Data Management

Overview

Menu	System administration → Archiving → Data management
Transaction code	arccfg
Function authorization	arccfg.*

Usage

You may use this application in order to view or edit the centrally managed settings for [archiving](#).

Integration

The settings for archiving/for data management are made centrally from all components/functions in the application.

Field descriptions

Product

In the Product field, the HYDRA module for which a rule is to be defined is entered.

Object

The Object defines what is to be provided .

Retention period

In the "retention period" fields, you define how long the data should be available before being archived.

Unit

The unit for the retention period is indicated in days, months or years.

Last retention date

This field is computed and subsequently indicates the last day on which data are still available in this object.

Action

It is possible to select from three process variants for an object:

D = delete	Object is deleted
M = move (archive)	Object will be transferred to the next division
X = Export	Object is unloaded (XUNLOAD format), subsequently deleted

Target object

At present, this automatically results from the detail configuration.

Condition

The characters in this field are added to the database command controlling the action, whereby this condition is linked to the conditions of the original command there and is set in parenthesis.

Last run (date/time)

Indicates when this rule was applied last.

License

Indicates which license is required for archiving. Multiple licenses may be entered here (separated by a space). If none of these licenses is licensed on the system, the data are either deleted (action D) or unloaded in files (action M or X) according to the relevant action.

Path

Optional path for generating the file export (Unload). At present, only local drives are supported on the HYDRA server in the archive. If no path is set, file exports are filed as follows:

<HYDRADIR>/<SYSTEM>/custom/archive/<YYYY-MM-DD>/<PRODUCT>/

HYDRADIR ... HYDRA directory

SYSTEM ... System number

YYYY-MM-DD ... Archiving date (YYYY ... Year, MM ... Month, DD ... Day)

PRODUCT ... Product from archiving configuration

Management table

Table name of management table where the archiving logs are stored.

Retention period management table

In the fields used for the Retention period, you define how long the logs should be available in the management table before being deleted.

Unit

The unit for the retention period is indicated in days, months or years.

Archiving step

Indicates whether this archiving executes archiving function I or II. In archiving function I, the data are transferred from the online inventory to the archive tables and/or deleted. Setting M (medium-term archive) In archiving function II, the data are transferred from the archive table to the file export and/or deleted. Setting L (long-term archive).

Configuration

Control indicator whether or not the configuration is active. Possible values: Y/N.

Archiving type

Identifier for time or object-related archiving. Supported modes: O = Object-related archiving (i.e. data are archived for each object individually). Z = Time-related archiving (i.e. data are collected/archived without any object reference). Please note: Time and object-related archiving differ from each other significantly in the archiving performance (runtime). Object-related archiving of mass data is not recommendable.

Priority

Integral value greater than 0. Indicates the execution sequence when several objects are defined to a module. Execution starts from the configuration with the lowest value.

Master table

Table including the archiving data. The extensions entered in the Condition field refer to this table.

Date column

Date column in the master table. Is used for evaluating the data to be archived with regard to the retention period.

Key 1

Clear key column in the master table; may be used to identify the data to be archived. Up to 5 key columns may be defined for object-related archiving. Time-related archiving only supports one primary key in Key 1.

Keys 2 – 5

Additional optional key columns for object-related archiving.

Comment

Comment line for describing the archiving configuration.



When copying the archiving configuration, only the data management configuration is copied at present, but no any defined data records from the object details. At present, the data records from the object details must be copied to the new configuration by the user in the appropriate screen.

Module	Version	Description of archiving
BDE	8.1	here
CAQ	8.1	here
MDE	8.1	here
MPL	8.1	here

4 Reload Manager

Summary

Menu	System Administration → Archiving → Reload Manager
Transaction code	arcrlid
Function authorization	arcrlid arcrlid.export arcrlid.import

Utilization

The customer is responsible for backing up data in a data archive and for restoring data. When data is backed up it is transferred from the long-term data area to the archive data area and, as a result, it is stored in a separate file system. The customer can start the backup process using the Reload Manager.

When data, which is filed in the archive data area, is restored, it is transferred back (copied) to the long-term data area. The customer has to do this and bears the responsibility.

To be able to evaluate restored data using standard evaluations/reports, the data has to be transferred back (copied) to the reload data area.

For this reason, the Reload Manager enables the following functions:

1. Moving of exported data into the customer archive (the customer archive path has to be defined within the HYDRA path settings).
2. Loading of exported data into the reload area for evaluations

Integration

The Reload Manager is a central function that is used by many components or functions.

Selection Criteria

The application provides the following selection criteria:

Module

Reference to the product group which archived data belongs to.

Object type

Object type of the archived data.



The selection options “module” and “object type” are mandatory fields.

Toolbar

Export

The “EXPORT” button allows for the entries selected in the Reload Manager to be exported to the customer directory (all data records can be selected using the context menu of the right mouse button). Once the function has been started, the input dialog that opens requires the customer archive path to be entered.

Import

The “IMPORT” button allows for the selected entries to be loaded into the reload area, in order for the data to be again available for HYDRA evaluations/reports. Once the function has been started, the data loading mode and an optional path are requested. By indicating the path, it is possible to specify another storage location for the files, provided that the archived files have not been moved to the customer-specific archive using the Reload Manager.

There are three different ways to deal with reload data after they have been reloaded to prevent the reload data set from increasing excessively (→ no slow evaluations/reports):

The following modes are distinguished:

- Cyclic

Data is loaded to the reload area in the “cyclic” mode. Loaded data is automatically removed from the corresponding reload tables, once the retention period specified within archive settings has expired.

Please compare the configuration entry HYD / RELOADMANAGER of the data management function. When demo settings are used, data is removed from the reload area after 14 days.

When data is imported, the HYD_REL_MANAGEMENT.DELETE_DATE field is calculated subject to the configuration and entered in the reload management table.

- User-specific

Data is loaded to the reload area in the “user-specific” mode. The loaded data is automatically removed from the corresponding reload area, once the time specified for the user within the user-specific settings has expired. Exception: In case several users loaded identical data into the reload area, the corresponding retention period is automatically set to the maximum date.

When data is imported, the HYD_REL_MANAGEMENT.DELETE_DATE field is computed subject to the configuration and entered in the reload management table.

- Manual

Data is loaded in the “manual” mode to the reload area. The customer is responsible for deleting data from corresponding reload tables. However, identical data cannot be loaded several times in the “manual” mode (data can be loaded in the “user-specific” mode though).

In order for data to be transferred to the reload data area, the HYDRA server must be able to access these files. Otherwise, loading is cancelled with an error message.